MEHADI HASAN

SENIOR TECH LEAD · AI ENGINEER · FORWARD DEPLOYED ENGINEER

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15+ Years · Aviation Security · Government · Fintech · Healthtech · Enterprise *AI Engineering · Forward Deployed*

PROFESSIONAL SUMMARY

Senior Tech Lead and **AI Engineer** with **15+ years** building mission-critical systems for **aviation security**, **government**, **fintech**, **healthtech** and **enterprise** clients. Works as a **Forward Deployed Engineer**: on site with the client, turning real operational problems into production systems. Currently **Sr. Tech Lead at Eleware Technology**, leading core modules of KLIA's airport security platforms (**PSIM, VHMS, AOP**), and in parallel leading 3 Malaysia Government transport programmes (LRT3, RTS, Smart Car Parking) at **Guildford Integrated System (GIS)**. Builds production **GenAI & Agentic AI**: **LLM integration** (OpenAI, Claude, Gemini, Ollama), **RAG**, **LangChain / LangGraph**, vector databases and **LLM fine-tuning with LoRA / QLoRA** (PEFT, bitsandbytes; own NeoHorse-Coder model) on **AWS Bedrock** and **Azure OpenAI**. Deep expertise in **system design (HLD/LLD)**, microservices, **event-driven architecture** (Kafka, RabbitMQ), **infrastructure as code with Ansible**, **n8n** automation and **MLOps / LLMOps**. Pursuing a **PhD in Computer Science**.

CORE COMPETENCIES

AI Engineering · **Forward Deployed Eng.** · **LLM Integration** · **Agentic AI** · **LoRA / QLoRA** · **Fine-Tuning** · **RAG** · **Vector Search** · **LangChain** · **LangGraph** · **LlamaIndex** · **AWS Bedrock** · **Azure OpenAI** · **Vertex AI** · **MCP Servers** · **Prompt Engineering** · **LLM Evals** · **MLOps / LLMOps** · **n8n** · **HLD/LLD** · **Microservices** · **Event-Driven** · **Kafka** · **RabbitMQ** · **Real-time** · **Cloud Native** · **CI/CD** · **Ansible** · **Airport Security / PSIM** · **Fintech** · **Healthtech** · **Blockchain** · **Carbon Credit / ESG** · **OWASP Top 10** · **AI Security** · **Open Source** · **Team Leadership**

TECHNICAL SKILLS

AI / GenAI Stack: OpenAI, Claude, Gemini, Grok, Mistral; Ollama, vLLM, llama.cpp; LangChain, LangGraph, LlamaIndex, Semantic Kernel; RAG, MCP, Function Calling, ReAct, AutoGen; AWS Bedrock, Azure OpenAI, Vertex AI; LangSmith, Langfuse, RAGAS, Promptfoo

LLM Fine-Tuning: LoRA, QLoRA (4-bit NF4), PEFT, Hugging Face Transformers, bitsandbytes, MLX, llama.cpp GGUF quantisation

Vector Databases: Pinecone, Weaviate, Qdrant, Chroma, FAISS, Milvus, pgvector

Programming Languages: Python, JavaScript/TypeScript, Java, C/C++, C#, PHP, Dart, Kotlin, Go

Frontend: React.js, Angular, Vue.js, Flutter, WPF, HTML5/CSS3

Backend: Node.js, NestJS, .NET 9/10, ASP.NET Core, ABP, Laravel, Spring Boot, FastAPI; REST, GraphQL, gRPC, SOAP

AI Pair Programming: Claude Code, GitHub Copilot, Gemini Code Assist, OpenAI Codex

Architecture: System Design (HLD/LLD), Microservices, Event-Driven, Domain-Driven Design, SOLID

Automation: n8n, Workflow Orchestration, AI-Driven Architecture, API Integration Patterns

Databases: PostgreSQL, MySQL, MongoDB, Redis, MSSQL, Oracle, SQLite, Firebase

Cloud / DevOps: AWS (Bedrock, EC2, S3, Lambda), Azure (OpenAI), GCP (Vertex AI), Docker, Kubernetes, **Ansible**, CI/CD, VMware ESXi, Grafana

Messaging & Real-time: Kafka, RabbitMQ, WebSocket, MQTT, SignalR, JMS

Methodology & Tools: Agile/Scrum, TDD, Redmine, Jira, Git/Bitbucket, Jenkins, Neovim, LaTeX, Overleaf

Custom-Built Tools: MetaSig, KafkaTUI, RabbitMQTUI, IPScanner (custom network mapping & visualization), custom ticket monitoring

Security Systems: PSIM, Milestone XProtect VMS, Security Expert access control, UVSS, SenseTime face recognition, LDAP / Kerberos

Networking & VPN: Tailscale, OpenVPN, WireGuard, multicast tracing, SignalR

PROFESSIONAL EXPERIENCE

Senior Tech Lead — Sep 2026 – Present

Eleware Technology Sdn. Bhd. – Airport Security Systems (KLIA) · Malaysia

- Lead core modules of 3 enterprise **airport security systems** at **KLIA**: **PSIM** (Physical Security Information Management), **VHMS** (Visitor & Vehicle Handling Management) and **AOP** (airport pass & permit platform)
- **PSIM:** one control-room console for alarms with SOP workflows, camera & door floor-plan maps, live/recorded video, cardholder tracking and server health – **.NET 9**, **ABP**, **Angular 20**, **SignalR** real-time push
- Integrate **Milestone XProtect** video, **Security Expert** access control, **under-vehicle scanners (UVSS)** and **SenseTime AI** face & appearance search; multi-node middleware with leased jobs for high availability
- **VHMS / AOP:** visitor & vehicle inspection, card printing, blacklist screening, e-vetting and AI face-quality checks on **.NET 10 / ABP**; mobile verifier app (.NET MAUI → Flutter)
- Production on **DC/DR** with **SQL Server Always On**; scripted deploy, rollback & health-check kits (PowerShell / SQL) for safe releases

Senior Software Engineer & Lead — Sep 2024 – Present

Guildford Integrated System (GIS) – Government Projects: LRT3, RTS, Smart Car Parking · Malaysia

- Lead engineer across **3 simultaneous Malaysia Government programmes**; own end-to-end HLD/LLD for **4-team, 18-engineer** organisation

- **LRT3:** C# .NET + PostgreSQL signalling backend processing **>2M events/day**; cut alarm noise ~**60%** via custom IP Scanner with multicast tracing
- **Smart Car Parking (TNG/Genting):** Flutter/Dart + NestJS + AI OCR plate detection; reduced gate latency from **4s** → **800ms** on integrated TNG/Mifare payment flow
- **MyDesk:** Custom AnyDesk-alternative unifying remote control of all 3 programmes; eliminated 3 commercial-licence costs (~**RM 60k/yr saved**)
- Introduced **LLM-assisted log triage** (Claude/OpenAI) + **n8n** ops pipelines; cut mean-time-to-diagnose by ~**40%**
- Built **Kafka TUI**, **RabbitMQ TUI** now standard diagnostic for LRT3 ops team; SCADA via JMS; CISCO VMware ESXi; **Grafana** monitoring

Engineering Lead (Contract, Remote) **May 2025 – Oct 2025**
Slotgator iGaming Project · Romania

- Delivered iGaming platform on **AWS + Kubernetes** sustaining ~**5k concurrent sessions**; zero P0 incidents over 6-month engagement
- Drove **Kafka** streaming + microservices, AES-256 payment hardening; led 7-engineer distributed team across 4 time zones

Full Stack Developer & DevOps Engineer **Dec 2023 – Sep 2024**
Gaming Industry · Malaysia & USA

- Shipped **4 hybrid iOS/Android apps** reaching **>100k installs**; managed **AWS EC2**, Tencent & Alibaba Cloud infra
- Integrated 3 payment rails (Decoin, Alipay, UnionPay) with AES-256 + real-time WebSocket; CI/CD cut release cycle from **2 weeks → 2 days**

Project Manager / Team Lead **Dec 2022 – Oct 2023**
Creative IT Institute · Chittagong

- Led **12-engineer** cross-functional team across 5 client products using Agile/Scrum; delivered **100% on-schedule** release rate
- Designed PostgreSQL/MySQL schemas powering ~**30k DAU**; standardised Git, VPS & Nginx operating procedures; mentored **15+** junior engineers

Team Development Lead & Sr. Full Stack **Mar 2022 – Oct 2022**
Datahostbd · Dhaka

- Mentored **8 junior engineers**; authored microservices training manual still in use; architected microservices with API gateway + service discovery (SOLID), reducing cross-service latency ~**35%** vs. monolith baseline

Senior Consultant & Full Stack Developer **Mar 2020 – Jan 2022**
Creative IT Institute · Chittagong

- Hardened 6 production codebases through code review + unit-test rollout; reduced post-release defects ~**50% YoY**; coordinated deployments serving ~**8k students**

Chief Consultant & Engineer **Jan 2017 – Aug 2019**
TRZ · Gazipur

- Built end-to-end ERP for garments industry from scratch; RFID-integrated tracking across **4 factory floors**, **600+ daily operators**
- Automated scheduled power management saving ~**18%** electricity cost; BI model in Pandas/NumPy/Matplotlib adopted by exec leadership for daily KPI review

CEO & Chief Consultant **Jul 2011 – Nov 2016**
Orimpo Inc. Ltd · Dhaka

- Founded & ran consultancy for 5+ years; delivered **40+ enterprise projects** on J2EE/JSP/Tomcat/J2ME across government, fintech, retail
- Led **bKash mobile payment prototyping** with Robi Axiata (Java) – foundational work for Bangladesh's largest mobile-money platform
- Directed engineering team; conducted security & network penetration assessments for govt & enterprise clients

SELECTED PROJECTS & OPEN-SOURCE

- **MetaSig – Cross-Platform Remote Device Management Platform** *(2025 – Present)* – Production fleet-management TUI for Android, iOS, macOS, Windows, Linux: WebSocket signalling, remote shell, file browser, live screen preview, APK

packaging & signing pipeline. Manages **>200 endpoints** across 3 Malaysia Gov programmes from a single terminal. *Stack:* Node.js, neo-blessed, WebSocket, Android agent

- **NeoHorse-Coder-4B – Custom Fine-Tuned Coding LLM (QLoRA)** *(Sep 2026 – Present)* – Own offline patch / fix / code-review model for local AI agents (OpenCode + custom model router), built over **35+ training iterations** on a 4B hybrid linear-attention base (Qwen3.5 family). **QLoRA** 4-bit NF4 (PEFT + bitsandbytes, r=32, α=64) on **5.4k** curated examples from own commits, CommitPackFT & agent traces; trained on RunPod A40 & AWS L4 GPUs; exported to GGUF (llama.cpp, Q4_K_M). Held-out eval vs base: grounded agent edits **10%** → **91%**, right-tool use **34%** → **94%**; speculative decoding **15** → **45 tok/s**; v34 test-writing specialist. *Stack:* PyTorch, Transformers, PEFT, MLX, llama.cpp
- **GPT-5.3 Agent Bridge + Pentest-AI Red-Team Deployment** *(2025 – 2026)* – Built OpenAI-compatible HTTP bridge wrapping ChatGPT GPT-5.3 via Playwright; integrated as multi-model agent inside **opencode** TUI for agentic dev workflows. *Separately:* authorised internal red-team deployment of upstream open-source Pentest-AI (ptai, original by 0xSteph) for OWASP Top 10 exercises via MCP – *integration role, not authorship*
- **Smart Car Parking System (TNG / Genting)** *(Sep 2024 – Present)* – AI OCR plate detection + C/C++ hardware gate control + TNG/Mifare payment integration; cut gate-pass latency from **4s** → **800ms**. *Stack:* Flutter/Dart, NestJS, C# WinForms, Python OCR, PostgreSQL
- **LRT3 / RTS Signalling + KafkaTUI / RabbitMQ-TUI** *(Sep 2024 – Present)* – Mission-critical Malaysia Gov. transport signalling processing **>2M events/day** (C#/.NET, PHP, Java, Grafana); cut alarm noise ~**60%**. KafkaTUI / RabbitMQ-TUI now standard diagnostic tools for LRT3 ops team
- **Medical AI Assistant (MSc Thesis) + AI Doctor Triage Platform** *(2024 – 2025)* – Healthcare AI with Claude / OpenAI & self-hosted Ollama; **RAG** over clinical KB (LangChain, LangGraph, Pinecone/Qdrant/pgvector, FastAPI, HIPAA-aware). Companion **AI Doctor** triage app: Gemini-powered, React Native mobile + Node.js backend + Prisma, Dockerised
- **Open-Source Suite – 16+ public repos** *(2023 – 2026)* – mydesk (AnyDesk-like remote control), macscp (Swift/AppKit SFTP), snmp-alarm (SNMP trap daemon), SignalR→WS gateway, nginx-RTMP for macOS, SuperCharge SG 2026, Free LLM API Keys, Math Practice App, 6 Redmine plugins (forum-pdf, doc-pdf, turbo-nav, issue-emailer, dark-mode, custom-mail)
- **Earlier flagship work** *(2015 – 2024)* – P2P Remote Desktop (C++/Qt, WebRTC, H.264/VP8); Custom Dev TUI Suite (create-project-tui, audio-tui-sox, db-exporter, debmaker); Job-Search AI TUI (9 sources, AI scoring & cover letters); n8n / Ollama self-hosted LLM infra; Telegram Dating Bot; Kids Education App (TF/Flutter); DMP crime clustering; RFID ERP + encrypted IRC for Bangladesh Army; UK e-commerce platform

EDUCATION

Doctor of Philosophy (PhD) – Computer Science — **Jul 2025 – Ongoing**
Lincoln University College · Part-time

Master of Computer Science — **Sep 2023 – May 2025**
Lincoln University College · Part-time – *Thesis: AI-Driven Medical Assistant*

BSc in Computer Science & Engineering — **2009 – 2017**
United International University · Full-time – *Project: An Intelligent Building Power Management System (IoT)*

Coursework: Java, PHP, Python, .Net, Ruby, J2EE, Embedded Systems, ML, Neural Networks, Data Mining, Statistical Inference. Awarded certifications in Programming Contest & Robotics.

CERTIFICATIONS & AWARDS

- Microsoft Imagine Cup 2012 – World's Premier Student Tech Competition
- Programming Contest Winner – Qazi Azher Ali Memorial, 2011
- UIU Robotics Workshop Certification, 2011
- NYDSA – National Youth Development Certificate
- Code Warriors Challenge – Programming Competition

ADDITIONAL CONSULTANCY

- **Alamin Sweaters** *(Feb – May 2013)* – Production workflow optimization, barcode tracking
- **Friends Appeals** *(Mar – Aug 2018)* – Business management software, DB architecture, reporting
- **Peregrine Air Line (UK)** *(Jan – Jun 2018)* – E-commerce, stock management, payment gateway